

SSD_Flash master production

1. Spinand master production

Because SPINAND has the property of bad blocks, it cannot be directly downloaded from the Flash Dump and then burned. You need to use the tools provided by the public version, and perform the generation according to the following steps.

1.1. Get the Mastering Kit

Click to download: [Master Film Making Tool](#) [media/SpinandPackBinTool_P2.rar]

1.2. Introduction to basic tool files

1. Executable program SstarMakeBin (SstarMakeBin_dual_env)
2. Initialization file SPINAND.INI(spinand)

The nandid in SPINAND.INI needs to be filled in according to the actual flash datasheet, `image_directory` which corresponds to the image path that needs to be packaged:

```

1  [path]
2  root_directory=./
3  image_directory=images/
4  script_file=auto_update_spinand.txt
5  outpath=./
6
7  [nand]
8  nandtype=SPINAND
9  nandid=0xC2-0x12
10
11 [env]
12 env_type=NANDRAW
13 env_part=ENV
14 #env_volume=MPOOL
15 #env_offset=0
16
17 [cis]
18 ecctype=10
19 nni=SPINANDINFO.sni
20 pni=PARTINFO.pni
21 #ppm=PAIRPAGEMAP_v2.ppm
22
23 [fcie]
24 type=fcie5
25
26 [boot]
27 #boottype=BFN
28 #miu=bfm_miu.bin

```

3. The configuration file of the burner is leap.def & snchk.def (leap burner uses leap.def, other burners generally use snchk.def)
4. The image files used to make the master film and the corresponding burning scripts are placed in the images folder:

images	2020/3/18 11:05	文件夹	
auto_update_spinand.txt	2020/3/4 14:36	文本文档	1 KB
readme.txt	2020/3/4 15:20	文本文档	4 KB
SPINAND.INI	2020/3/18 11:08	配置设置	1 KB
SstarMakeBin	2020/3/4 12:43	文件	2,049 KB
leap.def	2020/3/17 20:53	DEF 文件	1 KB
snchk.def	2020/3/17 20:53	DEF 文件	1 KB

The images folder is `project\image\output\images` copied from the images below

5. Script to configure partition and image paths `auto_update_spinand.txt`
`auto_update_spinand.txt` The partition information in the line under dynpart must be consistent with the mtdparts under uboot:

```

SigmaStar #
SigmaStar # print
baudrate=115200
bootargs=console=ttyS0,115200 ubi.mtd=UBI,2048 root=ubi:rootfs rw rootfstype=ubifs init=/linuxrc rootwait=1 LX_MEM=0x7f00000 mma_heap=mma_heap_name0,miu=0,sz=0x3000000 mma_memblock
mtd0:384k@128k(IPL0),384k(IPL1),384k(IPL_CUST0),384k(IPL_CUST1),768k(UBOOT0),768k(UBOOT1),384k(ENV0),0x60000(LOG0),0x500000(KERNEL),0x500000(RECOVERY),-(UBI)
bootcmd=ifconfig on; mw 1f001cc0 11; gpio out 8 0; nand read.e 0x22000000 KERNEL 0x500000; gpio out 8 1; bootm 0x22000000;nand read.e 0x22000000 RECOVERY 0x500000; bootm 0x22000000
bootdelay=0
bootparts=mtdparts=nand0:0x140000@0x0(UNKNOWN0),0x60000(IPL0),0x60000(IPL1),0x60000(IPL_CUST0),0x60000(IPL_CUST1),0xC0000(UBOOT0),0xC0000(UBOOT1),0x60000(ENV0),0x60000(BOOTLOG0),0x
00(RECOVERY),0x7100000(UBI)
ethact=sstar smac
ethaddr=00:73:14:00:00:01
fileaddr=23af42e8
filesize=1fb
gatewayip=172.19.24.1
initrd_block=77
initrd_high=0x21000000
initrd_size=67beed
ipaddr=192.168.1.20
mtddevname=IPL0
mtddevnum=0
mtdsize=nand0=nand0
mtdparts=mtdparts=nand0:384k@128k(IPL0),384k(IPL1),384k(IPL_CUST0),384k(IPL_CUST1),768k(UBOOT0),768k(UBOOT1),384k(ENV0),0x60000(LOG0),0x500000(KERNEL),0x500000(RECOVERY),-(UBI)
netmask=255.255.255.0
partition=nand0,0

```

6. For the ubi file format, extra space is required to store the ubi information, so the ubi partition cannot be used up, and the following space needs to be reserved for storing the ubi information:

Common 1G bit, 2G bit, 4G bit NAND reserves are calculated as 9MB, 17MB, 34MB

Otherwise, the following error of insufficient peb will be reported when making the master film:

```

last_partition : nand0,0
env_partition : nand0,0
mtdparts_init 1764 enter
Creating dynamic volume appconfigs of size 5242880
[ubi_create_volume]: search for vacant volume ID
[ubi_create_volume]: not enough PEBs, total=903, avail=40, vol reserved=42
UBI error: ubi_create_volume: cannot create volume 3, error -28
[do_mstar]: command error!
[do_mstar]: command error!
[do_mstar]: command error!
[nandbin:1319]update fail, please check log
[main]: Make spinand bin failed

```

Eg: The flash of the public version of spinand 128M, the total size of the ubi partition is 903 blocks, and the size to be reserved is $\geq 903 \times 2 \times 4k + 4 \times 4k$

```

Checksum ok!!
IDB:      StartBlk:      BlkCnt:      BackupBlkCnt:      PartType:
0:      0,(0000000000) 10,(000140000) 0,(0000000000) 0X0020,(CIS)
1:      10,(0X00140000) 1,(0X00020000) 2,(0X00040000) 0X0003,(IPL0)
2:      13,(0X001A0000) 1,(0X00020000) 2,(0X00040000) 0X0003,(IPL1)
3:      16,(0X00200000) 1,(0X00020000) 2,(0X00040000) 0X0001,(IPL_CUST0)
4:      19,(0X00260000) 1,(0X00020000) 2,(0X00040000) 0X0001,(IPL_CUST1)
5:      22,(0X002C0000) 2,(0X00040000) 4,(0X00080000) 0X0006,(UBOOT0)
6:      28,(0X00380000) 2,(0X00040000) 4,(0X00080000) 0X0006,(UBOOT1)
7:      34,(0X00440000) 3,(0X00060000) 0,(0000000000) 0X000D,(ENV)
8:      37,(0X004A0000) 1,(0X00020000) 0,(0000000000) 0XF000,(KEY_CUST)
9:      38,(0X004C0000) 3,(0X00060000) 0,(0000000000) 0X0002,(LOG0)
10:     41,(0X00520000) 40,(0X00500000) 0,(0000000000) 0X0012,(KERNEL)
11:     81,(0X00A20000) 40,(0X00500000) 0,(0000000000) 0X0009,(RECOVERY)
12:    121,(0X00F20000) 903,(0X070E0000) 0,(0000000000) 0X0021,(UBI)

```

So in `project\image\configs\i2m\spinand.ubifs.p2.partition.config`, the total size of the ubi partition cannot exceed `0X070E0000 - 903*2*4k + 4*4k`

```

66 rootfs$(RESOURCE) := $(OUTPUTDIR)/rootfs
67 rootfs$(FSTYPE) := ubifs
68 rootfs$(PARTSIZE) := 0xA00000
69 rootfs$(BOOTENV) := console=ttyS0,115200 ubi.mtd=UBI,2048 root=ubi:rootfs ro rootfstype=ubifs init=/linuxrc rootwait=1
70
71 miservice$(RESOURCE) := $(OUTPUTDIR)/miservice/config
72 miservice$(FSTYPE) := ubifs
73 miservice$(PARTSIZE) := 0xA00000
74 miservice$(MOUNTPT) := /config
75 miservice$(MOUNTPT) := ubi0:miservice
76 miservice$(OPTIONS) := rw
77 miservice$(OTABLK) := /dev/ubi0_1
78
79 customer$(RESOURCE) := $(OUTPUTDIR)/customer
80 customer$(FSTYPE) := ubifs
81 customer$(PARTSIZE) := 0x5000000
82 customer$(MOUNTPT) := /customer
83 customer$(MOUNTPT) := ubi0:customer
84 customer$(OPTIONS) := rw
85 customer$(OTABLK) := /dev/ubi0_2
86
87 appconfigs$(RESOURCE) := $(OUTPUTDIR)/appconfigs
88 appconfigs$(FSTYPE) := ubifs
89 appconfigs$(PARTSIZE) := 0x500000
90 appconfigs$(MOUNTPT) := /appconfigs
91 appconfigs$(MOUNTPT) := ubi0:appconfigs
92 appconfigs$(OPTIONS) := rw
93 appconfigs$(OTABLK) := /dev/ubi0_3

```

7. The default is to read the SPINANDINFO.sni & PARTINFO.pni files from the images directory, so before making the master, you need to copy the SPINANDINFO.sni & PARTINFO.pni under images\boot to the images, otherwise the following error will be reported:

```

nand_init_chip:41 enter
[_MDrv_SPINAND_GET_INFO]: Cannot open ./images/SPINANDINFO.sni
#####
#          SPINAND INFO          #
#####
[MDrv_SPINAND_Init]: page size: 2048
[MDrv_SPINAND_Init]: spare size: 64
[MDrv_SPINAND_Init]: page count: 64
[MDrv_SPINAND_Init]: block count: 1024
[board_spinand_init]: Cannot open pni file ./images/PARTINFO.pni
#####

```

1.3. Execute SstarMakeBin_dual_env (SstarMakeBin) to generate nand.bin

SstarMakeBin 跟 SstarMakeBin_dual_env 是相同作用的 bin，但是版本不一样，请注意使用版本，使用错误，可能会出现烧录后再 uboot 提示 ENV CRC ERROR。

在执行打包命令前，需要将 `SpinandPackBinTool\images\scripts_bin\` `[[set_partition.es` 下这行屏蔽掉：

```

1 # <- this is for comment / total file size must be less than 4KB
2 mtdparts del CIS
3 #setenv mtdparts $(mtdparts),0x60000(logo),0x500000(kernel),0x500000(recovery),-(ubi)
4 saveenv
5 nand erase.part ubi
6 ubi part ubi
7 ubi create rootfs 0xA00000
8 ubi create miservice 0xA00000
9 ubi create customer 0x5100000
10 ubi create appconfigs 0x300000
11
12 % <- this is end of file symbol
13

```

然后在linux执行

1. 执行 `./SstarMakeBin_dual_env -n SPINAND.INI` (TAKOYAKI_DLS00V008 之后 (不包含) 使用)
2. 或者 `./SstarMakeBin -n SPINAND.INI` (TAKOYAKI_DLS00V008以前 (包含) 使用)

这样会生成一个nand.bin

```

aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTools$ ./SstarMakeBin -n SPINAND.INI
[parse_ini_file:1161]ini file=SPINAND.INI
[iniparser_line][467]: [path]
[iniparser_line][467]: root_directory=./
[iniparser_line][467]: image_directory=images/
[iniparser_line][467]: script_file=auto_update_spinand.txt
[iniparser_line][467]: outpath=./
[iniparser_line][467]:
[iniparser_line][467]: [nand]
[iniparser_line][467]: nandtype=SPINAND
[iniparser_line][467]: nandid=0xC2-0x12
[iniparser_line][467]:
[iniparser_line][467]: [env]
[iniparser_line][467]: env_type=NDRAW
[iniparser_line][467]: env_part=ENV
[iniparser_line][467]: #env_volume=MP00L
[iniparser_line][467]: #env_offset=0
[iniparser_line][467]:

```

```

Nand Flash basic information:
main=2048 bytes(Hex:0x800), spare=64 bytes(Hex:0x40)
main+spare=2112 bytes(Hex:0x840), page number per block=64
block size=0x20000(128KB, main), block size=0x21000(main+spare)

```

```

[nand truncate_outfile]: SPINAND type
[nand truncate_outfile]: truncate file with length 0x1b75000
[nandbin:1355]Nand bin file done!!

```

```

aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTools$
aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTools$ ls
auto_update_spinand.txt images leap.def nand.bin readme.txt snchk.def SPINAND.INI SstarMakeBin
aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTools$

```

这个bin就是烧录程序，然后使用烧录器烧录即可。

1.3. 制作母片出错检查

1. 如果是执行生成bin时有报错，先检查一下是否所执行的命令不支持，比如sd的升级脚本需要改为tftp的脚本形式
 2. 是否执行了一些save命令，重新保存了分区信息，导致分区信息与在auto_update_spinand.txt中指定的分区不一致
 3. 如果是烧录到板子上跑不起来，首先根据分区信息，检查生成的bin文件对应偏移地址处的文件信息是否正确，比如IPL的分区偏移为0x140000，则检查生成的bin文件地址0x140000位置处的文件内容是否为IPL文件的内容
 4. 检查烧录器烧录时oob区是否disable
 5. 用flash tool工具分别dump出板子上flash开始处、IPL分区偏移处、IPL_CUST分区偏移处、UBOOT分区偏移处、kernel分区偏移处的部分文件对比是否与烧录的bin文件一致
-

1.5. 支持的FLASH 列表

目前支持的flash列表参考[SSD_FLASH支持列表](#) [flash_p2.html]

1.6. 烧录器参考配置(分区配置文件)

1.6.1. snchk.def分区配置文件说明 (常用于非leap烧录器)

如下图：def格式是大端存储的

1. 使用二进制打开文件，每一行代表一个分区的描述
2. 红框中的分区描述：
 - a. 红框1中，4个字节描述分区的起始block，如第一行00 00 00 00，代表的是起始位置为第0个block
 - b. 红框2中，4个字节描述分区的结束block，如第一行09 00 00 00，代表的是结束位置为第9个block (包含第9block)
 - c. 红框3中，4个字节描述分区的实际有效数据占用的block数，如第一行0A 00 00 00，代表的是占用了10个block

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f	
00000000h:	00	00	00	00	09	00	00	00	0A	00	00	00	43	49	53	00	;CIS.
00000010h:	0A	00	00	00	0C	00	00	00	01	00	00	00	49	50	4C	30	;IPL0
00000020h:	0D	00	00	00	0F	00	00	00	01	00	00	00	49	50	4C	31	;IPL1
00000030h:	10	00	00	00	12	00	00	00	01	00	00	00	49	50	4C	5F	;IPL_
00000040h:	13	00	00	00	15	00	00	00	01	00	00	00	49	50	4C	5F	;IPL_
00000050h:	16	00	00	00	1B	00	00	00	02	00	00	00	55	42	4F	4F	;UBOO
00000060h:	1C	00	00	00	21	00	00	00	02	00	00	00	55	42	4F	4F	;!.....UBOO
00000070h:	22	00	00	00	24	00	00	00	01	00	00	00	45	4E	56	30	; "...\$......ENV0
00000080h:	25	00	00	00	25	00	00	00	01	00	00	00	4B	45	59	5F	; %...%......KEY_
00000090h:	26	00	00	00	28	00	00	00	01	00	00	00	4C	4F	47	4F	; &...(......LOGO
000000a0h:	29	00	00	00	50	00	00	00	11	00	00	00	4B	45	52	4E	;)...P.....KERN
000000b0h:	51	00	00	00	78	00	00	00	11	00	00	00	52	45	43	4F	; Q...x.....RECO
000000c0h:	79	00	00	00	FF	03	00	00	00	02	00	00	55	42	49	00	; y... ..UBI.
000000d0h:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	;

结束block – 起始block >= 有效数据block

1.6.2. leap.def分区配置文件说明（针对leap烧录器）

如下图：def格式是大端存储的

1. 除去第一行跟最后一行，每一行就是代表一个分区的描述
2. 第一行是leap的头信息，烧录器厂商可以自行判断是否需要，不需要可去掉（因烧录器而异）
3. 每行的前4个字节为特有标记，各烧录器厂家自行判断是否需要，不需要可去掉（因烧录器而异）
4. 红框中的分区描述：
 - a. 红框1中，4个字节描述分区的起始block，如第一行00 00 00 00，代表的是起始位置为第0个block
 - b. 红框2中，4个字节描述分区的结束block，如第一行09 00 00 00，代表的是结束位置为第9个block（包含第9block）
 - c. 红框3中，4个字节描述分区的实际有效数据占用的block数，如第一行0A 00 00 00，代表的是占用了10个block

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f	
00000000h:	47	52	4F	55	50	20	44	45	46	49	4E	45	32	00	00	00	; GROUP DEFINE2...
00000010h:	01	00	00	00	00	00	00	00	09	00	00	00	0A	00	00	00	;
00000020h:	01	00	00	00	0A	00	00	00	0C	00	00	00	01	00	00	00	;
00000030h:	01	00	00	00	0D	00	00	00	0F	00	00	00	01	00	00	00	;
00000040h:	01	00	00	00	10	00	00	00	12	00	00	00	01	00	00	00	;
00000050h:	01	00	00	00	13	00	00	00	15	00	00	00	01	00	00	00	;
00000060h:	01	00	00	00	16	00	00	00	1B	00	00	00	02	00	00	00	;
00000070h:	01	00	00	00	1C	00	00	00	21	00	00	00	02	00	00	00	;!
00000080h:	01	00	00	00	22	00	00	00	23	00	00	00	01	00	00	00	;"....#.....
00000090h:	01	00	00	00	24	00	00	00	25	00	00	00	01	00	00	00	;\$...%.....
000000a0h:	01	00	00	00	26	00	00	00	3B	00	00	00	12	00	00	00	;&...;.....
000000b0h:	01	00	00	00	3C	00	00	00	51	00	00	00	12	00	00	00	;<...Q.....
000000c0h:	01	00	00	00	52	00	00	00	CD	00	00	00	2E	00	00	00	;R...?.....
000000d0h:	01	00	00	00	CE	00	00	00	49	00	00	00	2E	00	00	00	;?..I.....
000000e0h:	01	00	00	00	4A	01	00	00	4A	01	00	00	01	00	00	00	;J...J.....
000000f0h:	01	00	00	00	4B	01	00	00	5F	01	00	00	15	00	00	00	;K..._.....
00000100h:	01	00	00	00	60	01	00	00	62	01	00	00	01	00	00	00	;`...b.....
00000110h:	01	00	00	00	63	01	00	00	63	01	00	00	01	00	00	00	;c...c.....
00000120h:	01	00	00	00	64	01	00	00	FF	03	00	00	7E	00	00	00	;d... ..~...
00000130h:	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	;

结束block – 起始block >= 有效数据block

1.6.3. 烧录器配置信息

- 应用产品信息

CPU品牌/型号	软件系统
Sigmastar	Linux

- 镜像文件中是否包含Spare(OOB)数据:

☐ 有Spare(OOB)数据: _____字节

☒ 无Spare(OOB)数据

- 坏块管理模式(是否有分区)

☐ 跳过方式: Skip IB

☒ 分区方式: Multiple Partitions (with skip IB)

需要提供分区表文件 (分区文件参考1.6.2说明)

- 是否使用On-chip ECC(仅带有片内ECC功能的Flash)

☐ 不使用On-chip ECC

最大允许错误位参数: _____(bits) / _____(bytes)

镜像文件中必须包含Spare(OOB)数据

√ 使用On-chip ECC

- 坏块标记

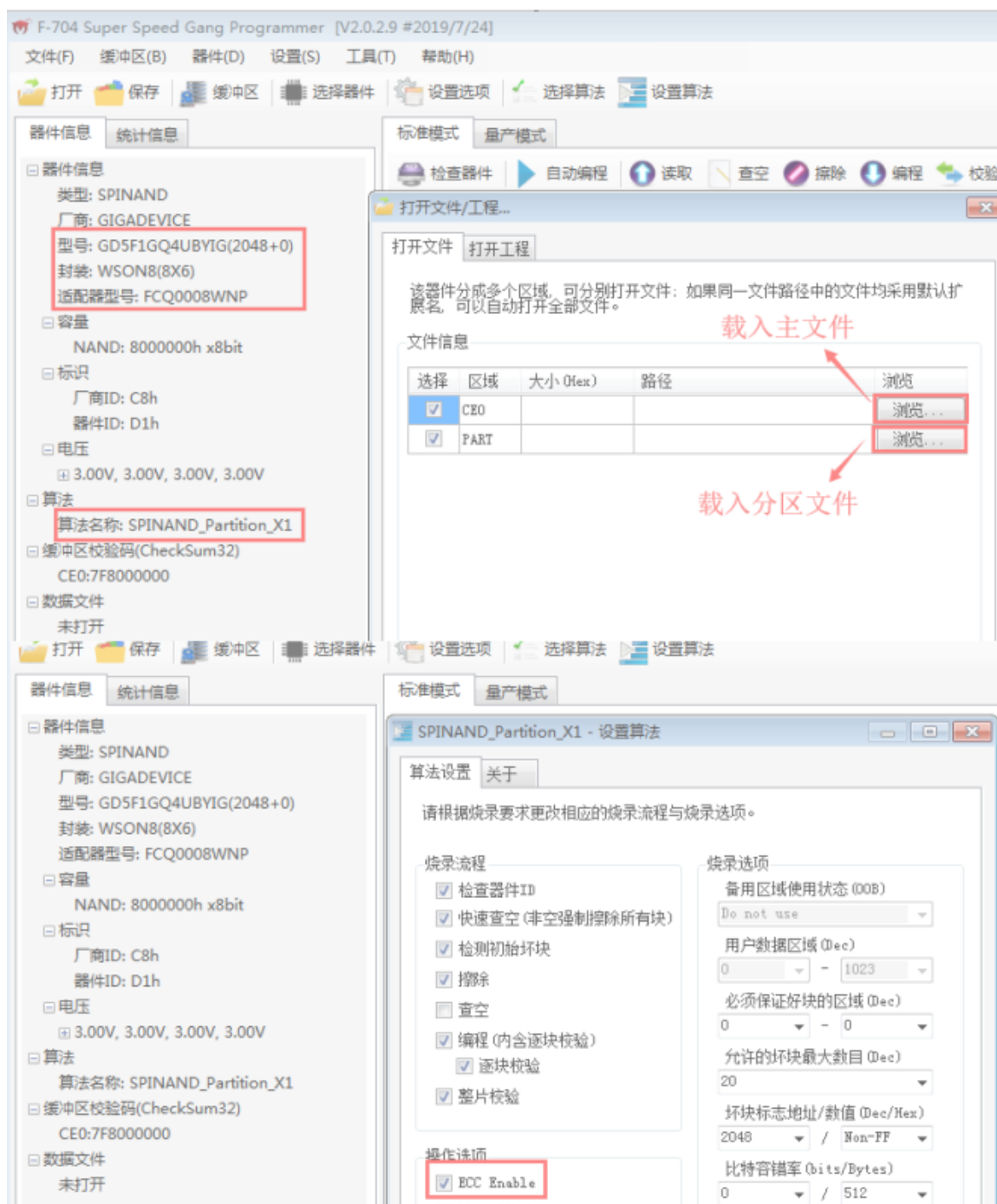
标记位置(地址): 800h

坏块标记数据: 00h

1.6.4. 烧录器配置参考

本章节内容仅供参考，不同烧录器配置可能不同。

使用针对GD5F1GQ4UBYG的flash配置，参考如下：



2. SPINOR母片制作

因为SPINOR Flash不存在坏块的情况，所以SPINOR Flash可以直接从一个确认完整的Flash中完整dump下来直接做二次烧录，母片制作较SPINAND简单很多，这里使用Flash_Tool去dump并烧录说明（实际使用过程中可自行使用烧录器也可）

2.1. 获取SSD对应的Flash_Tool.exe

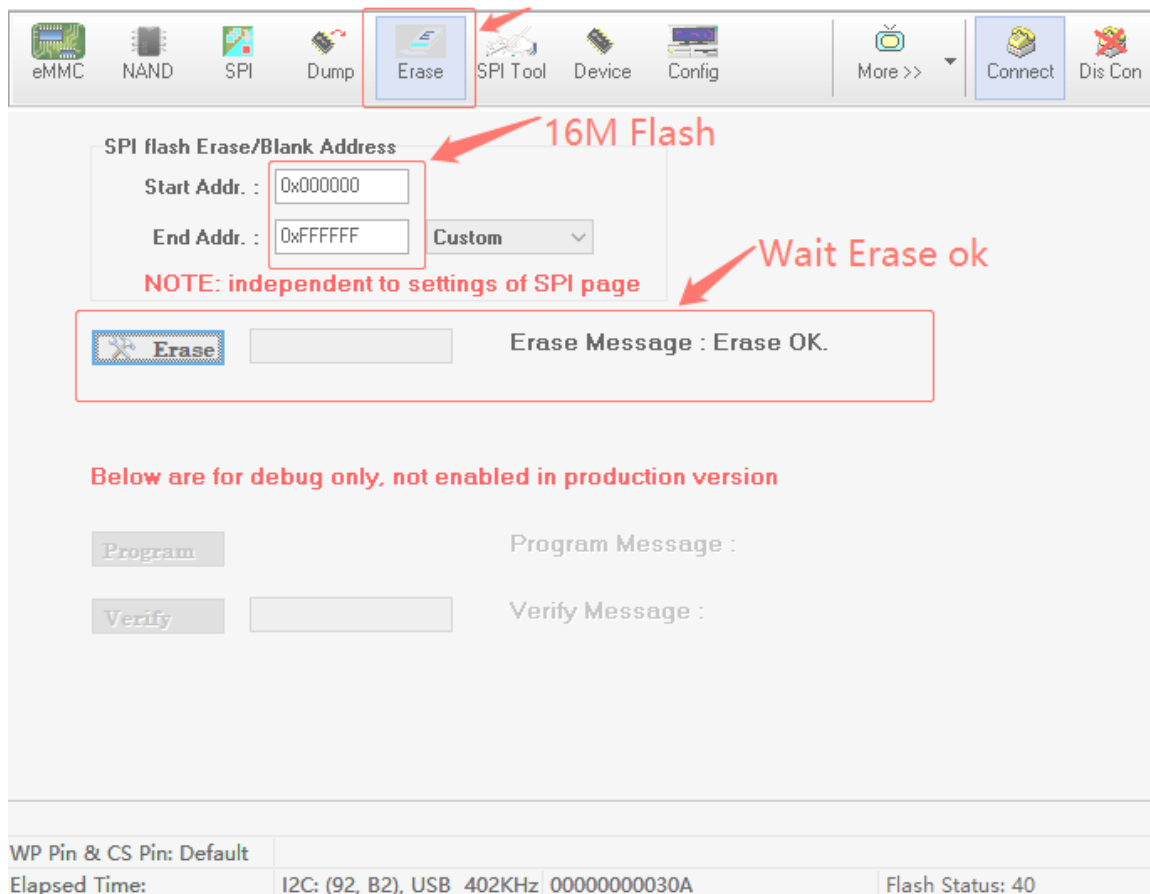
2.2. 准备一个干净的Flash

假设是在用的Flash，则接用Flash_Tool去做全chip擦除，擦除操作如下：

a. 开机进入uboot控制台（如果可以），在uboot控制台输入“debug”，并断开串口终端工具，如下图：

```
SigmaStar #  
SigmaStar # debug  
debug mode on, cmdline is disabled  
  
SigmaStar #  
Connection closed.  
  
Disconnected from remote host(COM4_串口) at 21:15:27.  
  
Type `help' to learn how to use Xshell prompt.  
[C:\~]$
```

b. 使用Flash_Tool擦除整个Flash，如下图Flash_Tool连接成功后切到Erase界面，填写好Start-End Addr擦除范围后（按全Flash大小算），点击Erase等待ok即可

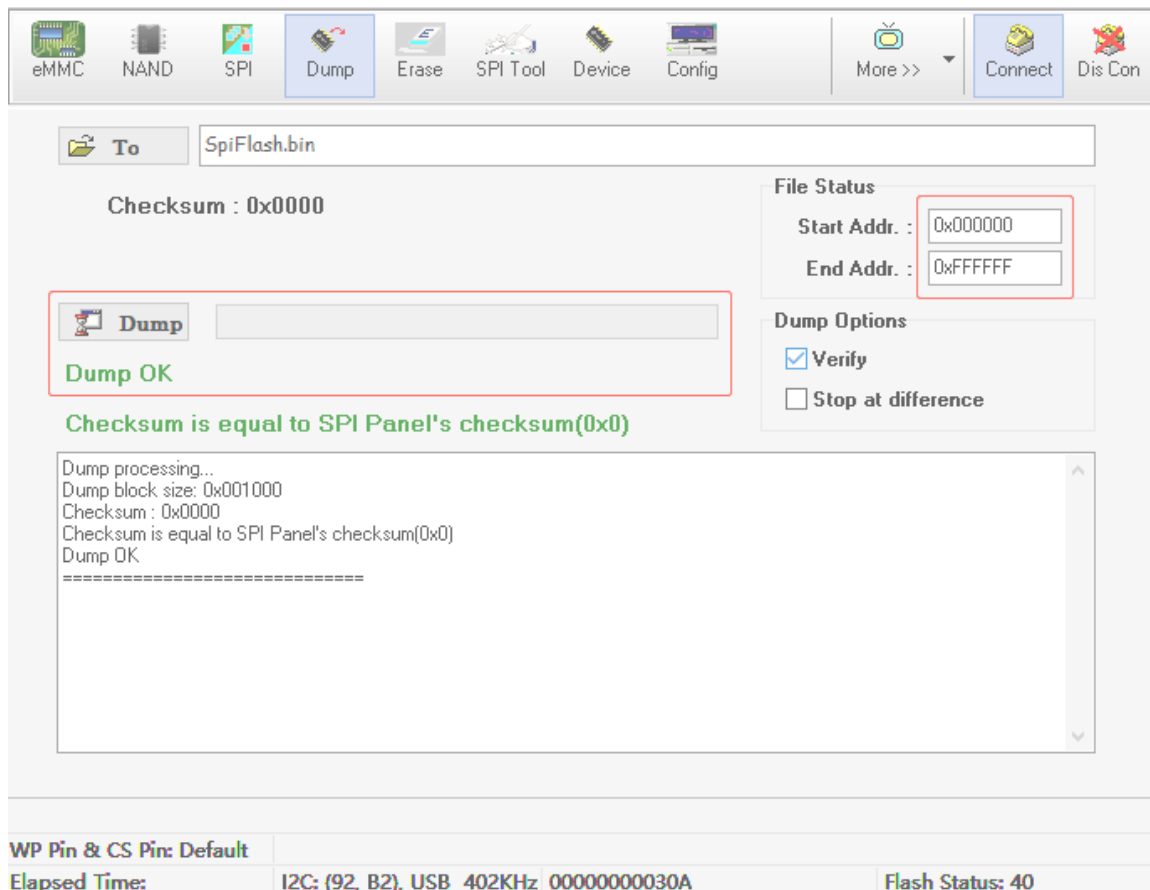


2.3. 烧录image

参照烧录文档，选择适合的方式将image烧录到干净的这块NorFlash上(烧录步骤这里省略，如有需要参考烧录相关文档即可)

2.4. Dump flash

使用Flash_Tool工具dump在干净状态烧录成功的Flash，生成的bin即可作为母片使用，dump操作如下，等待dump成功即可(注意dump的地址要填全flash范围)



2.5. 验证母片的正确性

验证母片的正确性，可以使用Flash_Tool从Flash的0地址按顺序烧录(烧录时间可能比较久)，烧录成功后开机看看是否正常即可

eMMC NAND SPI Dump Erase SPI Tool Device Config		More >>	Connect Dis Con
Image: C:\Users\bill.dong\Desktop\SpiFlash_test.bin			
Run << Less	Checksum : 0xF8C9		
Src: [2020/5/26 11:48:32]		Hex files Unused Bytes: ☒ 0x00 ☐ 0xFF	
Pass Erase OK. Program Message : Programming at 0xFF0000 Verify Message : Verifying at 0xFF0000 Verify : Each time 001000 Bytes Verify OK. [Important] Send the 0x31 & 0x01 command to change status as 0x00. Unblock Finished! Block Run Flow Done. End time: 12:31:32 [Important] Send the 0x31 & 0x01 command to change status as 0x00. Unblock Finished!		File Status Start Addr. : 0x000000 End Addr. : 0x00FFFFFF	
		☐ ReConnect ☐ Multi Flashes ☒ Block Run Flow ☒ Erase Device ☐ All Chip ☒ File Area ☐ Erase Area ☐ First 512 KBytes ☐ Partial Erase Setup ☐ Base shift at 0x00000000 Hot Key Ctrl+R ☐ Chip Reset After Finish ☐ Disable hot key SpiFlash_X ☐ Flash_0 ☐ Flash_1 ☐ Flash_2 ☐ Flash_3	
		☐ Blank ☐ HDCCP Key HDCCP... Key #:0 ☒ Program ☐ Verify ☒ Exit ISP Type: SPINOR	
WP Pin & CS Pin: Default			
Elapsed Time:		I2C: (92, B2), USB 402KHz 000000000112 Flash Status: 00	